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(54) STACKING TRAY

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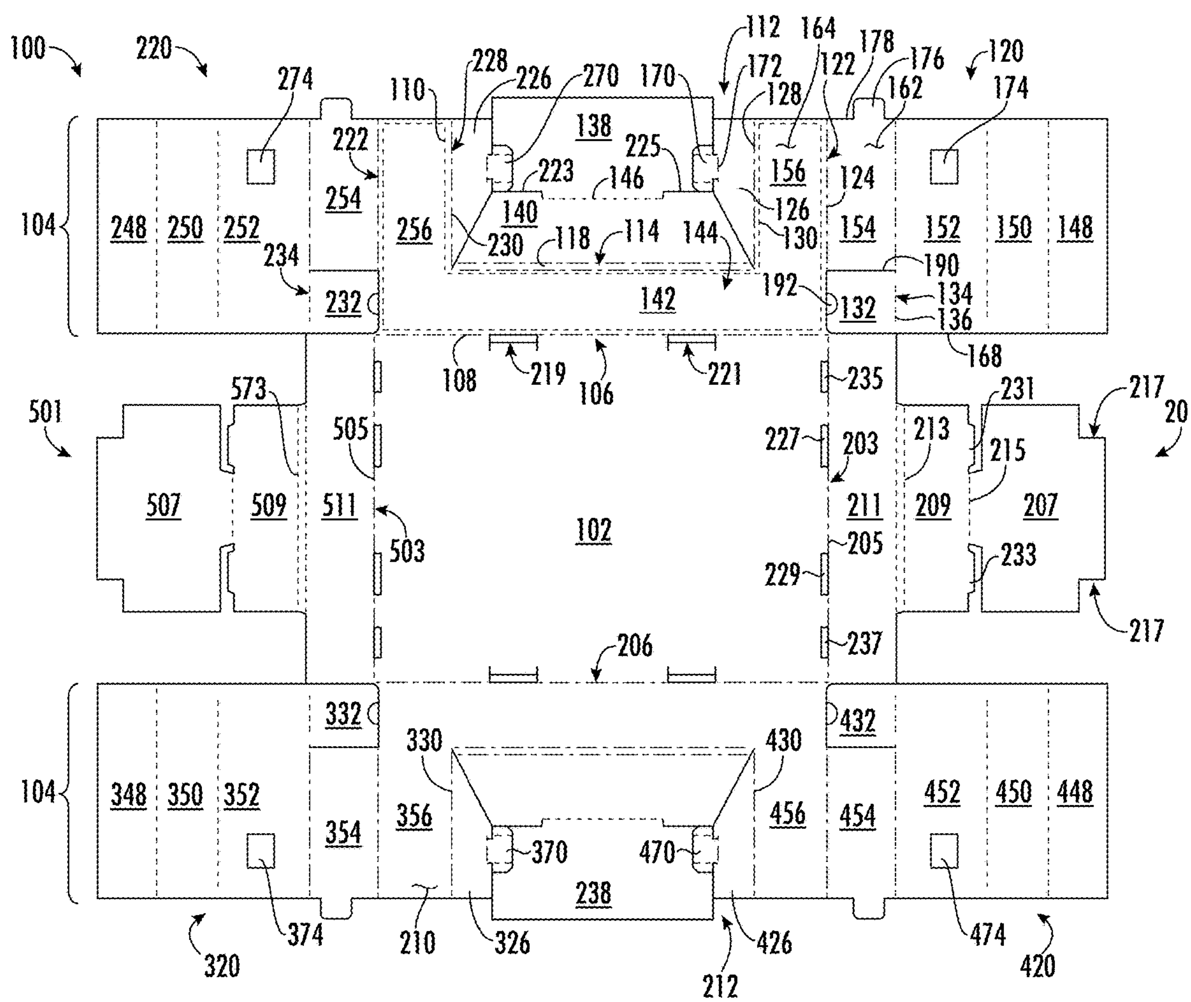
(57) **ABSTRACT**

In accordance with at least one aspect of this disclosure, a blank for forming a container includes a bottom panel and an end panel extending outwardly from a first edge of the bottom panel. The end panel includes a stacking pillar panel extending laterally from a first edge of the bottom panel, an end roll over panel extending laterally from a horizontal edge of the stacking pillar panel configured to roll over the stacking pillar panel such that a liner panel of the end roll over panel is configured to lay flat on the bottom panel in a set-up container, a corner roll over panel extending laterally from a first vertical edge of the stacking pillar panel configured to roll over itself onto the stacking pillar panel to form a corner stacking pillar, a locking flap extending a laterally from a second vertical edge of the stacking pillar panel configured to lock the corner stacking pillar together in the set-up container, and a retaining flap extending from an inner edge of the corner roll over panel configured to retain the corner stacking pillar in an upright position in the set-up container.



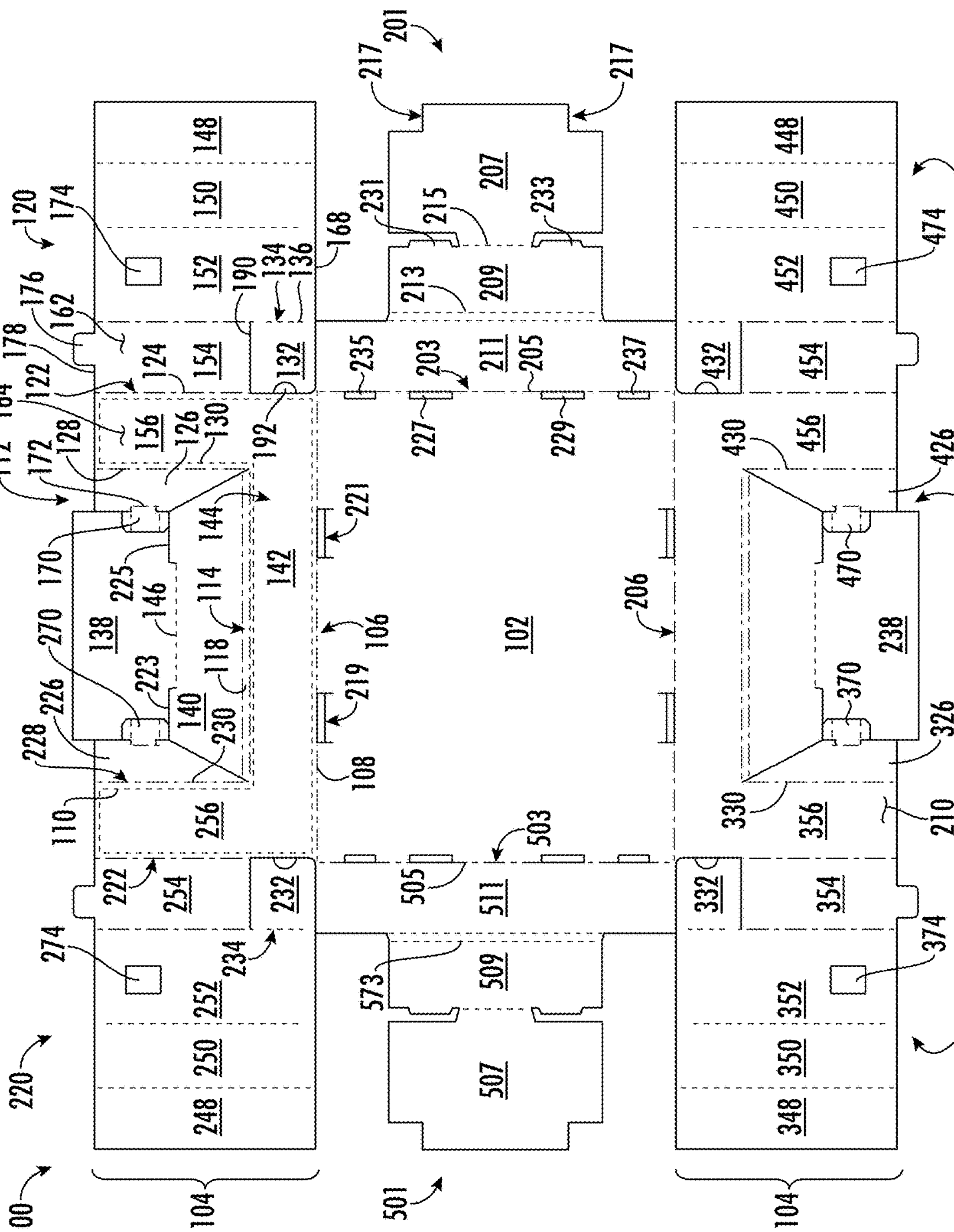


FIG. 1

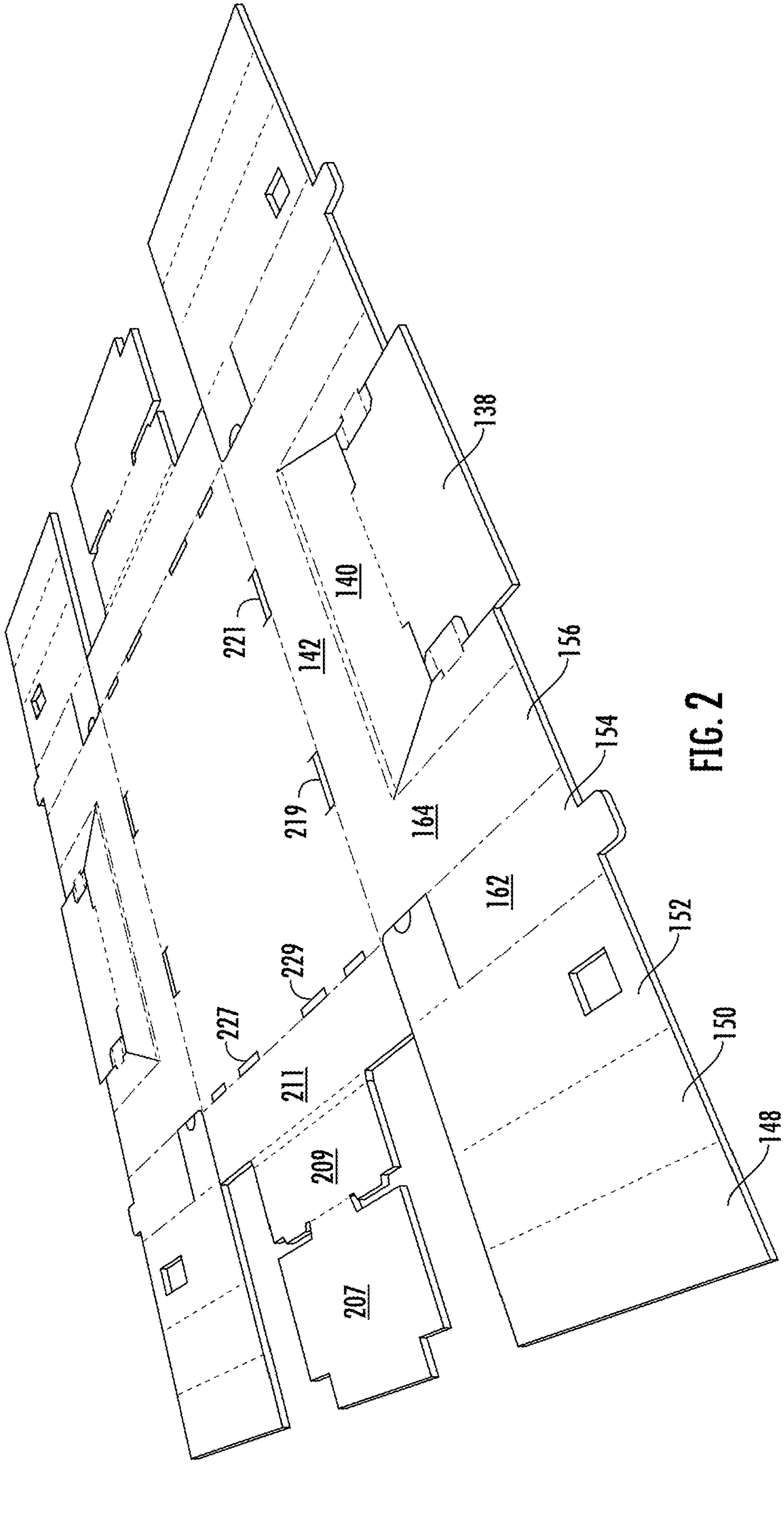

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FIG. 2

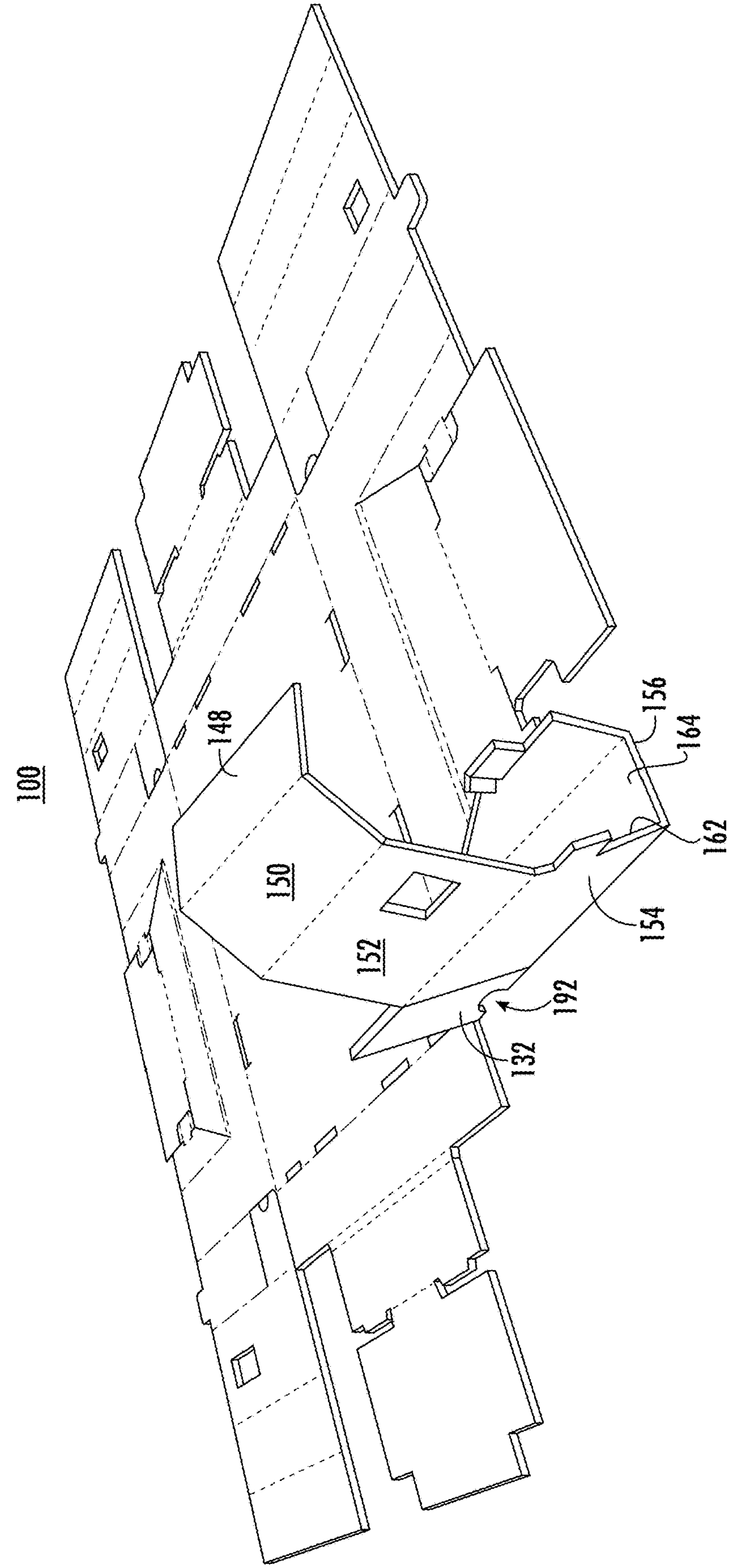
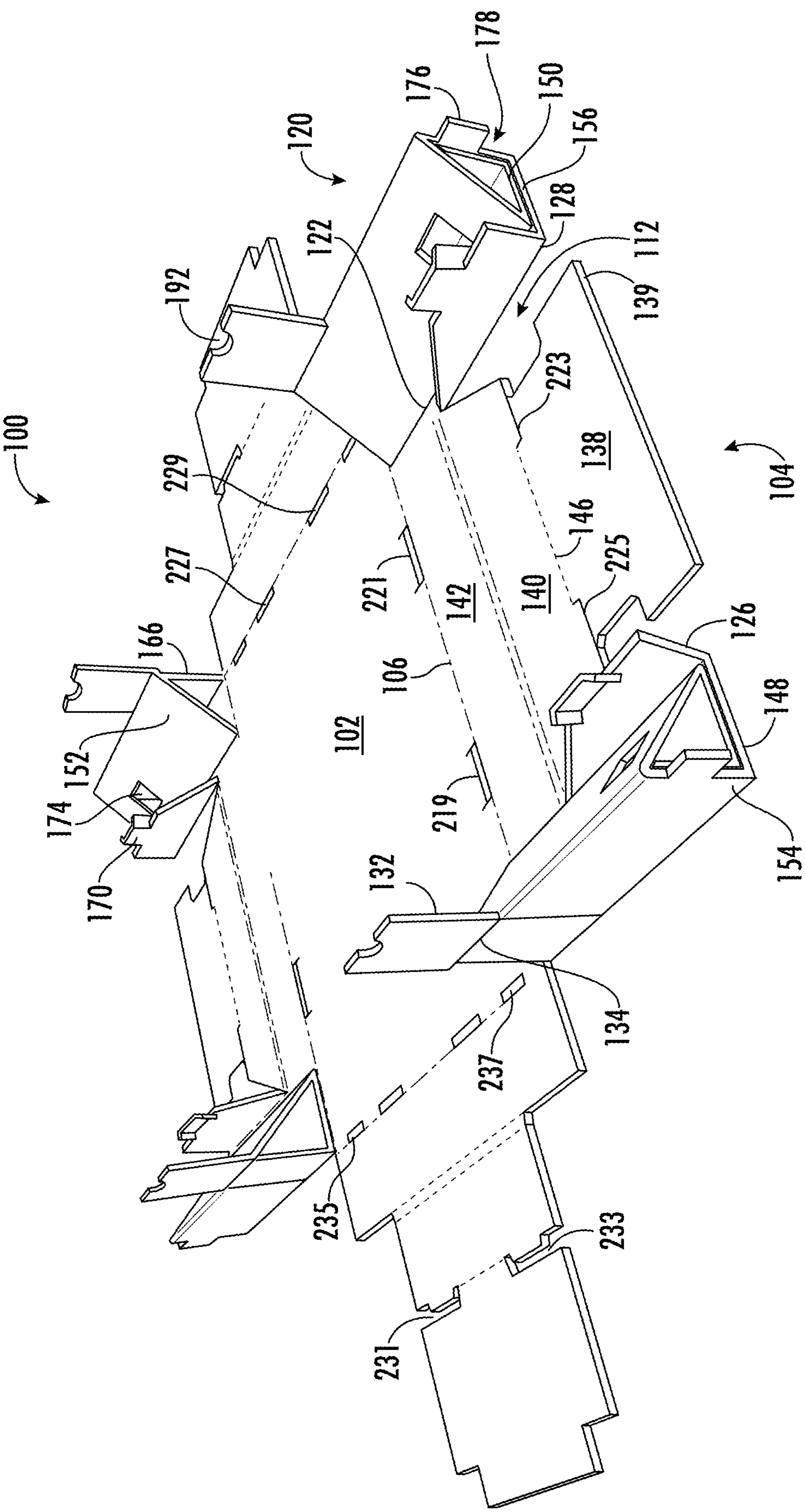


FIG. 3



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104
106
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FIG. 4

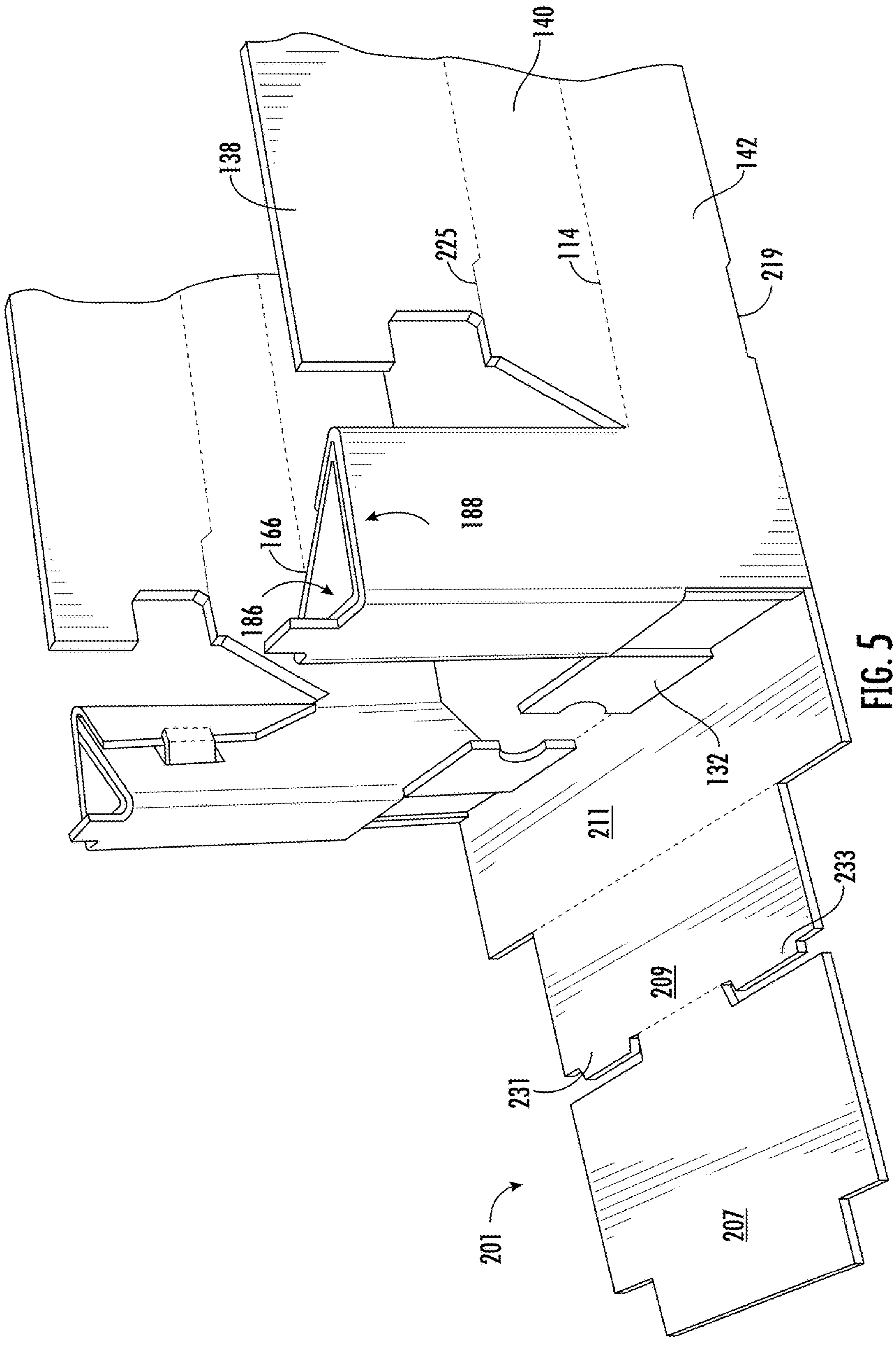


FIG. 5

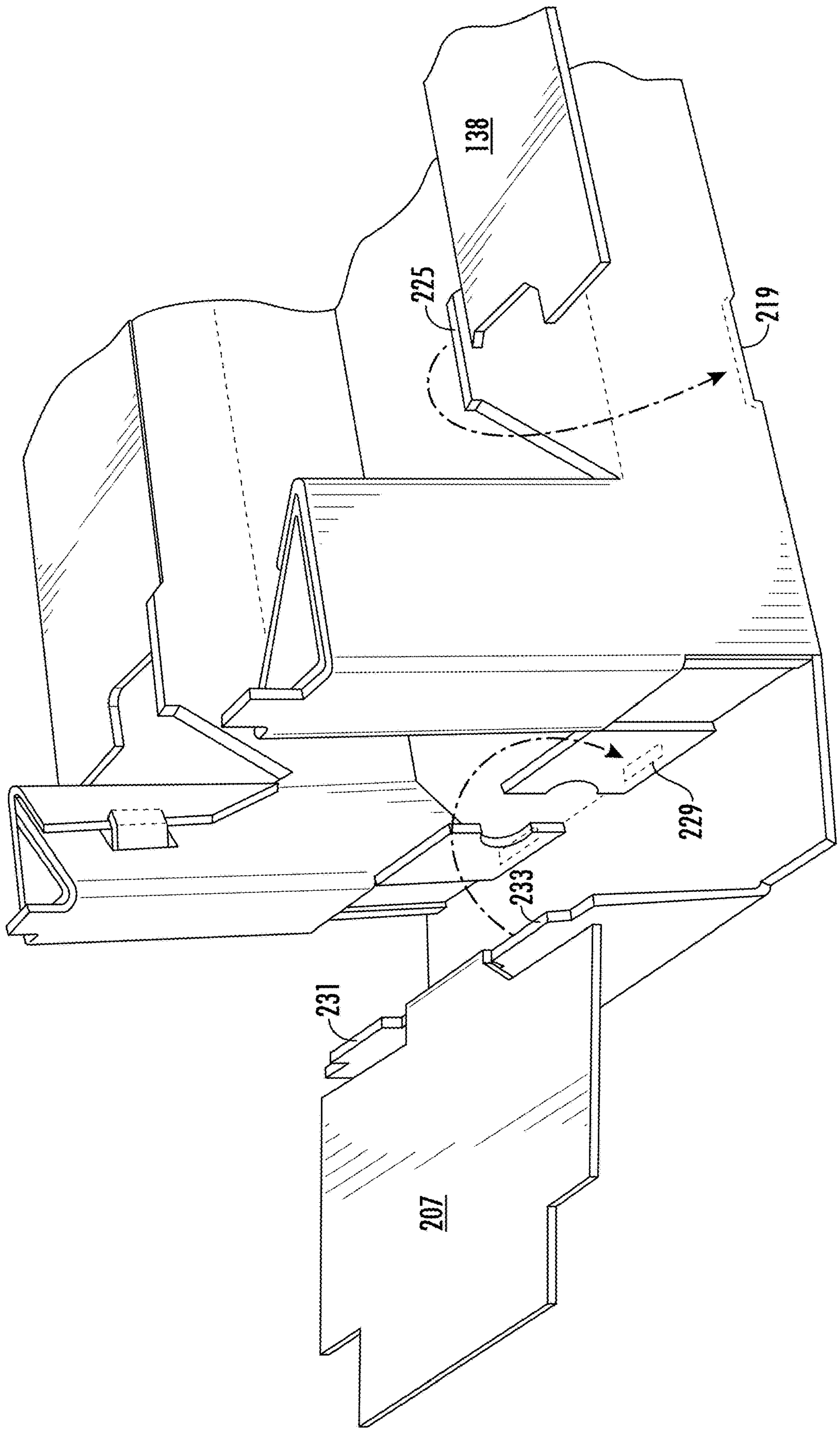
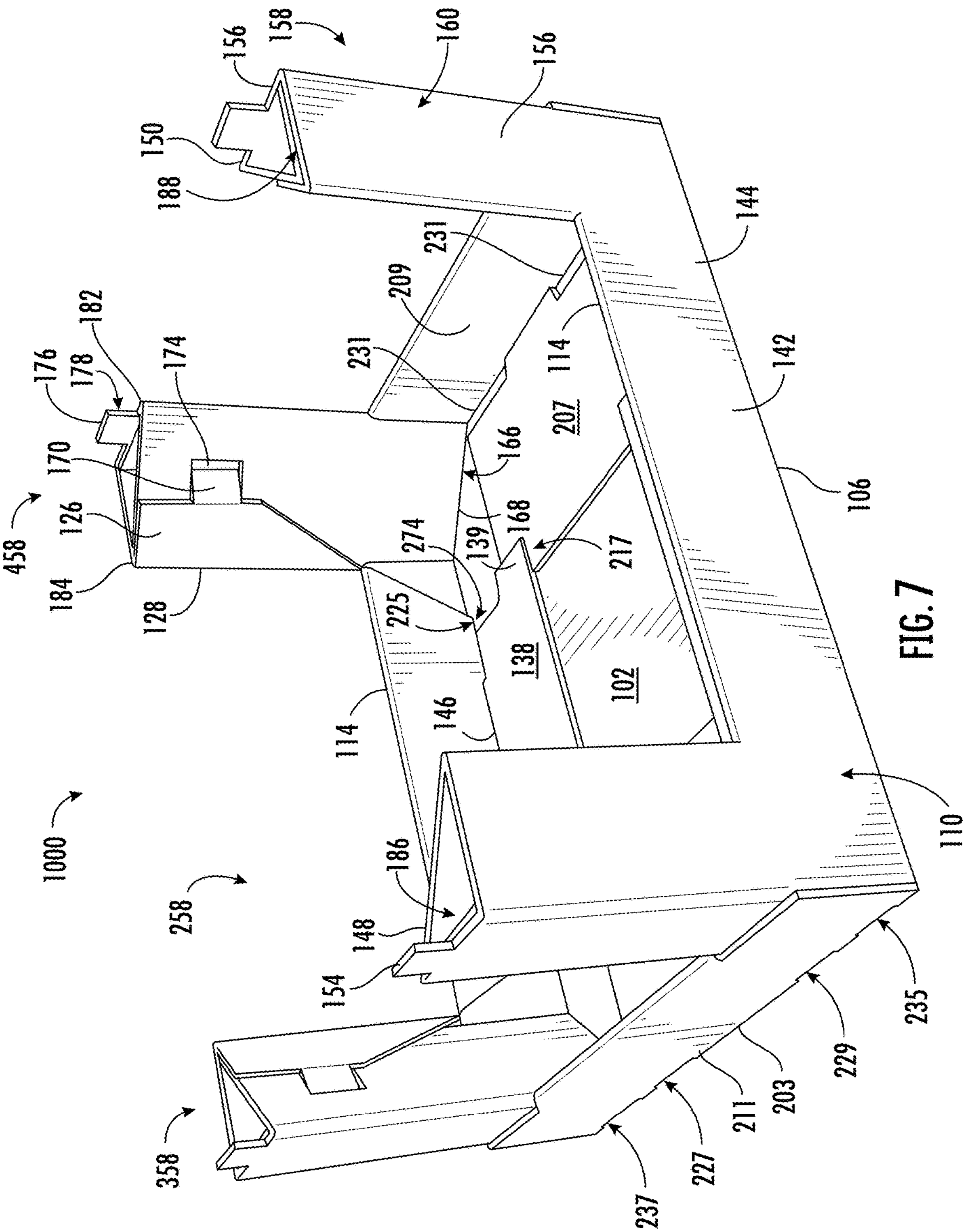


FIG. 6


158
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207
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217
106
274
139
184
128
225
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146
110
1000
186
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203
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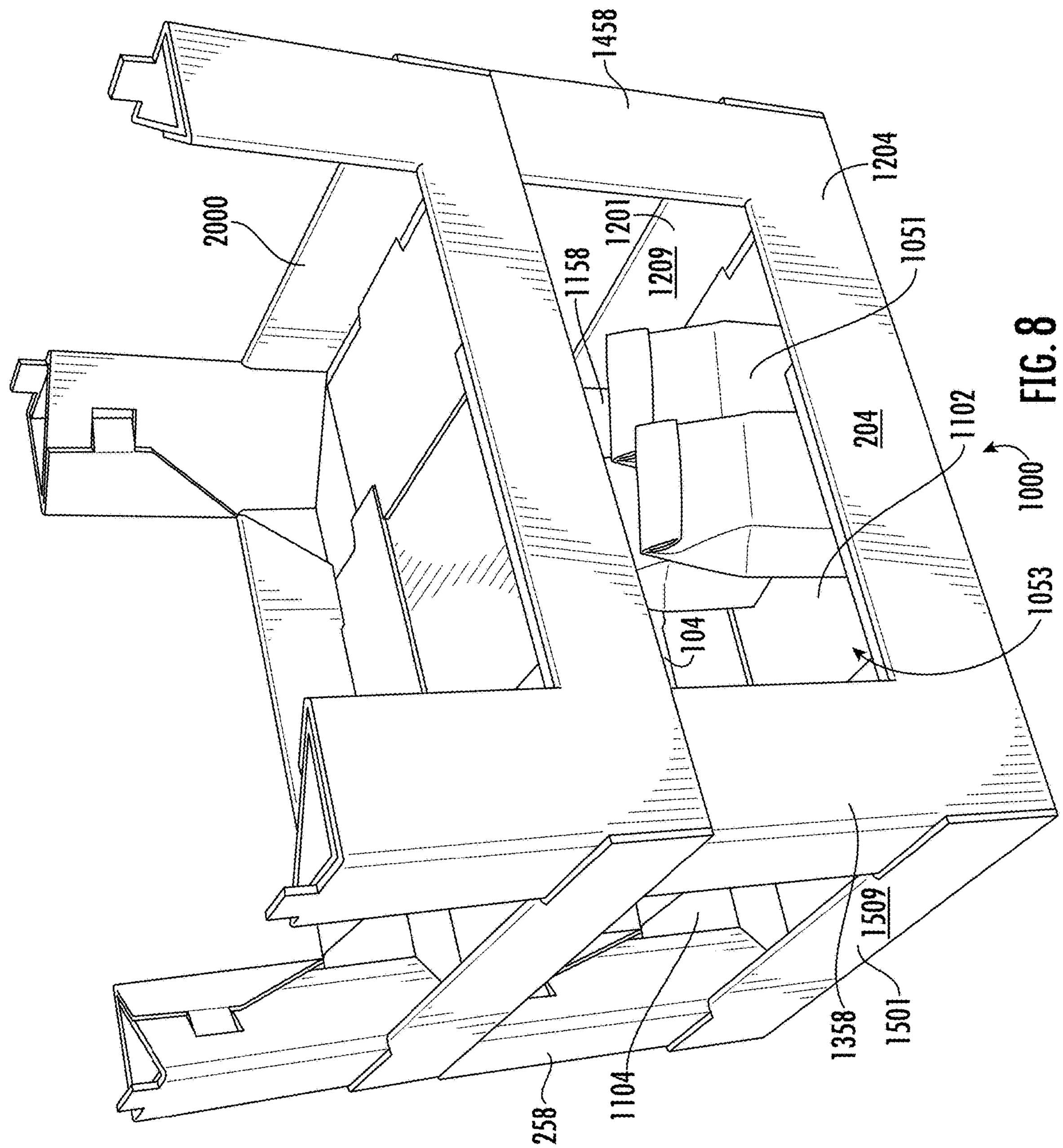
FIG. 7


1458
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2000
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FIG. 8

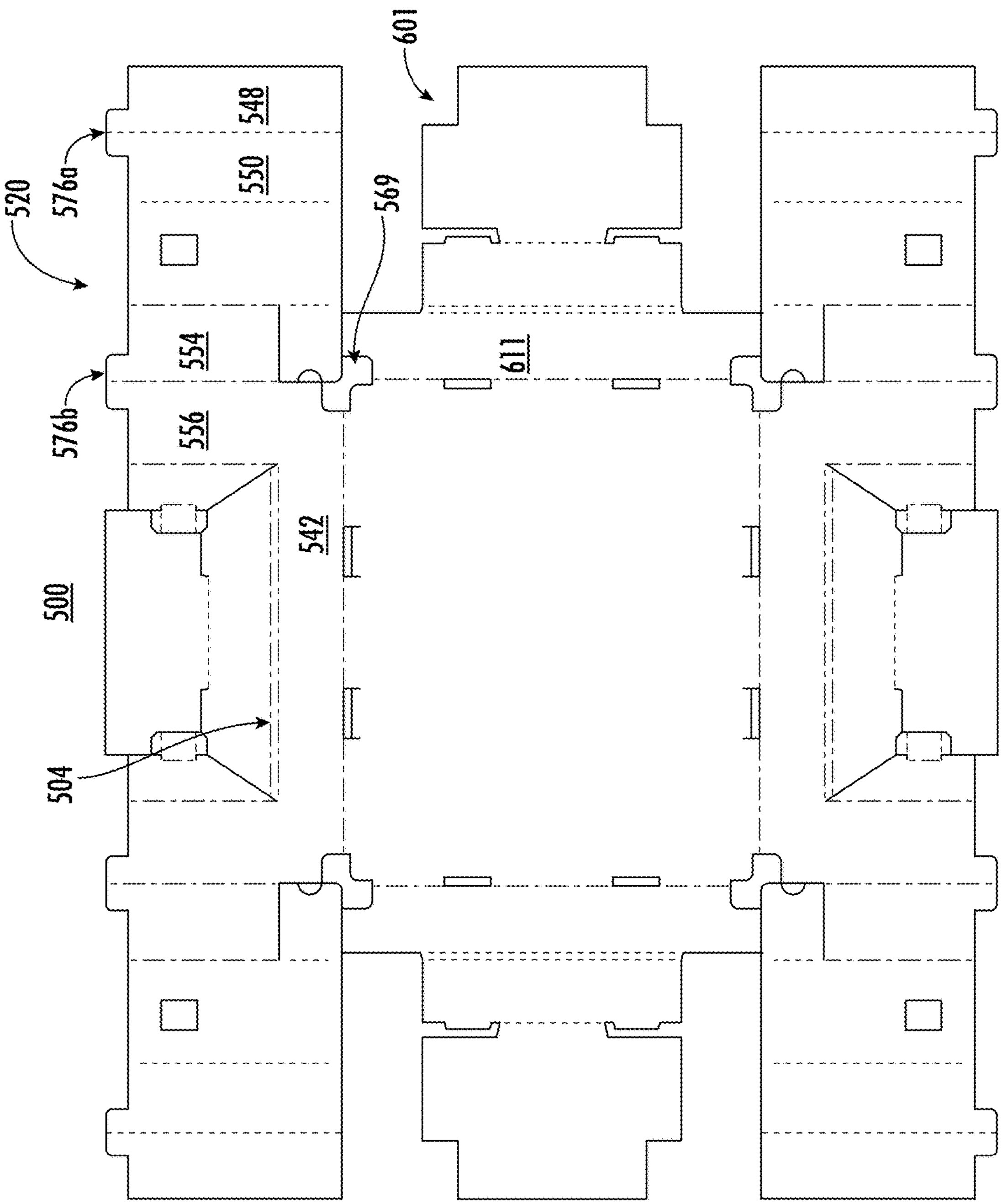

FIG. 9

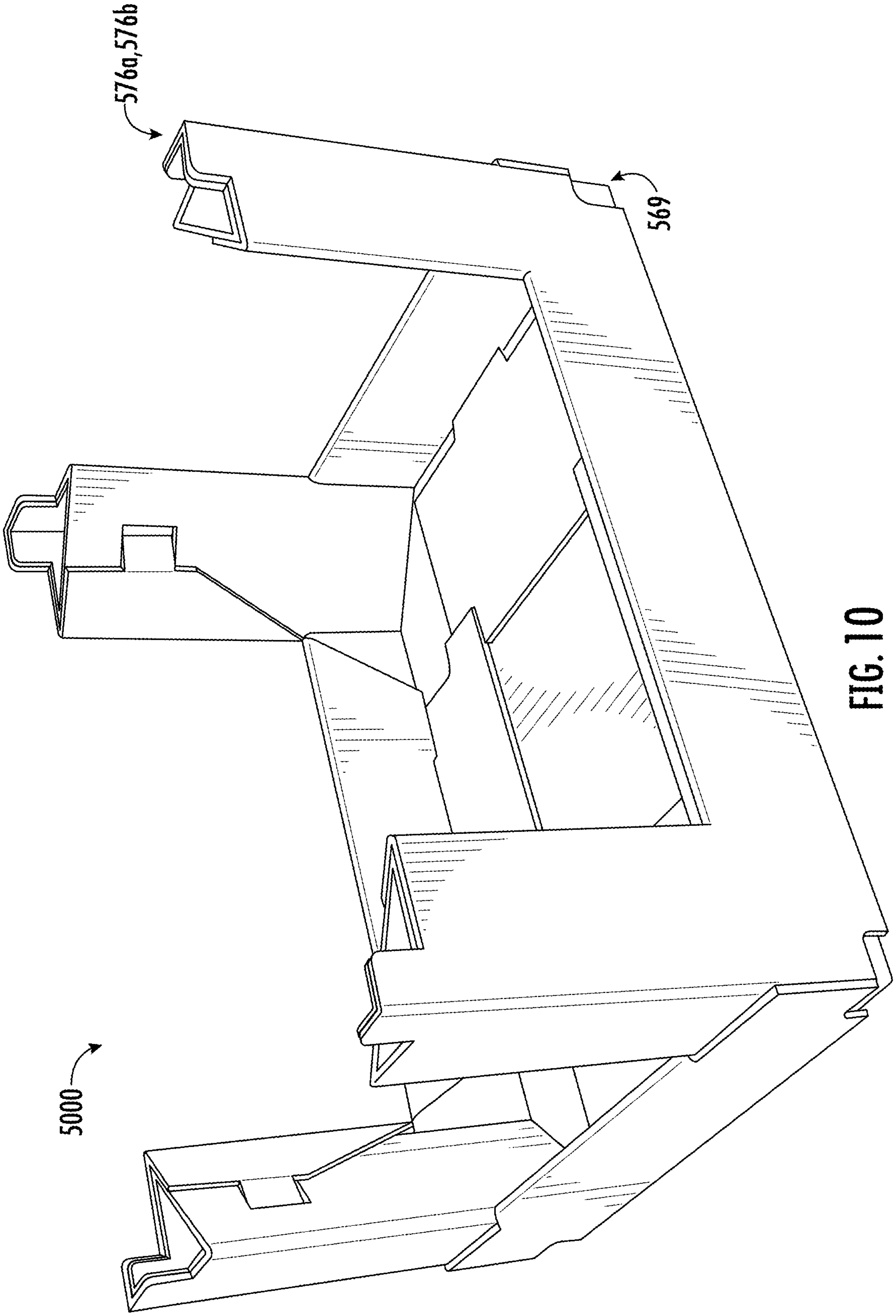

FIG. 10

STACKING TRAY

TECHNICAL FIELD

[0001] The present disclosure relates generally to a display tray, and specifically to a display tray having improved visibility and stacking abilities.

BACKGROUND

[0002] Pallet displays in Club Stores are beneficial since a large amount of product can be merchandised in one location without having to be re-stocked often. Many of these designs incorporate a full depth "X" or similar divider insert that separates the tray into quadrants and spans the corners, providing critical stacking support for the multiple tray layers, that is locked into a low-profile base tray. However, the "X" dividers tend to obstruct the customer's view of the product, particularly when the primary customer facing side of the display has been emptied and product in the other quadrants are difficult to see or cannot be seen by the customer because the insert blocks the view. These trays are hard to or impossible to rotate in order to show what product is located in the other quadrants. Further, the dividers cannot be removed without affecting the integrity of the tray stack.

[0003] There is still a need in the art for improved display trays having improved stacking abilities and visibility to each corner of the tray from multiple vantage points. The present disclosure provides a solution for this need.

SUMMARY

[0004] In accordance with at least one aspect of this disclosure, a blank for forming a container includes a bottom panel and an end panel extending outwardly from a first edge of the bottom panel. The end panel includes a stacking pillar panel extending laterally from a first edge of the bottom panel, an end roll over panel extending laterally from a horizontal edge of the stacking pillar panel configured to roll over the stacking pillar panel such that a liner panel of the end roll over panel is configured to lay flat on the bottom panel in a set-up container, a corner roll over panel extending laterally from a first vertical edge of the stacking pillar panel configured to roll over itself onto the stacking pillar panel to form a corner stacking pillar, a locking flap extending a laterally from a second vertical edge of the stacking pillar panel configured to lock the corner stacking pillar together in the set-up container, and a retaining flap extending from an inner edge of the corner roll over panel configured to retain the corner stacking pillar in an upright position in the set-up container.

[0005] The blank further includes a side roll over panel extending outwardly from a second edge of the bottom panel configured to roll over itself to such that the retaining flap of the corner roll over panel is sandwiched between an inner panel and an outer panel of the side roll over panel. A liner panel of the side roll over panel is configured to lay flat on the bottom panel abutting the liner panel of the end roll over panel in the set-up container. In embodiments, the stacking pillar can be a triangular stacking pillar.

[0006] The corner roll over panel includes, a first inner panel, a second inner panel, a medial panel, a first outer panel, and a second outer panel, wherein the first inner panel is configured to lay flush on an inside surface of the first outer panel, wherein the second inner panel is configured to lay flush on an inner surface of the second outer panel with the corner stacking pillar constructed, wherein the medial panel is configured to form a hypotenuse of the stacking pillar in a constructed stacking pillar.

[0007] The first inner panel and the first outer panel of the corner roll over panel can be configured to form a first two ply side of the constructed stacking pillar and the second inner panel and the second outer panel of the corner roll over panel can be configured to form a second two ply side of the constructed stacking pillar to reinforce a stacking strength of the constructed stacking pillar relative to a direction of gravity. A bottom edge of the medial panel of the corner roll over panel can be configured to lay flush on the bottom panel for an entire length of the bottom edge of the medial panel (e.g., the bottom edge is flat and continuous along the bottom panel for its length).

[0008] The medial panel of the corner roll over panel can include a locking aperture defined therein to receive a locking tab of the locking flap in a constructed triangular stacking pillar.

[0009] The first outer panel can include a stacking tab extending laterally from a top edge of the first outer panel configured to be inserted into a stacking aperture of another container stacked on top of the set-up container.

[0010] In certain embodiments, at least the first outer panel includes a stacking tab extending laterally from a top edge of the first outer panel configured to be inserted into a stacking aperture of another container stacked on top of the set-up container. In certain embodiments, the first outer panel, the second outer panel, the first inner panel, and the second inner panel can each define a portion of the stacking tab such that the stacking tab formed in the set up container is defined along a corner of the stacking pillar. In certain embodiments the stacking tab can be two-ply.

[0011] In embodiments, the bottom panel can include a first locking aperture and a second locking aperture defined in the first edge of the bottom panel, and the inner panel of the end roll over panel can include a first locking tab and a second locking tab configured to be inserted into the first locking aperture and the second locking aperture, respectively, of the bottom panel in the set-up container.

[0012] The bottom panel can also include a third locking aperture and a fourth locking aperture defined in the second edge of the bottom panel and the inner panel of the side roll over panel can include a first locking tab and a second locking tab configured to be inserted into the third locking aperture and the fourth locking aperture, respectively, of the bottom panel in the set-up container.

[0013] In embodiments, the bottom panel can further include a fifth locking aperture and a sixth locking aperture defined in the second edge of the bottom panel configured to receive a respective stacking tab of a second container when the set-up container is stacked atop the second container.

[0014] In embodiments, the corner roll over panel is a first roll over corner panel, and the end panel can further include a second corner roll over panel extending laterally from a third vertical edge of the stacking pillar panel, configured to roll over itself onto the stacking pillar panel to form a second corner stacking pillar, a second locking flap extending laterally from a fourth vertical edge of the stacking pillar panel configured to lock the second corner stacking pillar together in the set-up container, and a second retaining flap extending laterally from an inner edge of the second corner roll over panel configured to retain the second corner stacking pillar in an upright position in the set up container.

The side roll over panel can be configured to sandwich the second retaining panel between the inner panel and the outer panel of the side roll over panel in the set-up container.

[0015] In embodiments, the side roll over panel can be a first side roll over panel and the end panel can be a first end panel, and the blank can further include a second end panel extending outwardly from a third edge of the bottom panel. The second end panel can include a second stacking pillar panel extending outwardly from the third edge of the bottom panel, a second end roll over panel extending laterally from a horizontal edge of the second stacking pillar panel configured to roll over the stacking pillar panel such that a liner panel of the second end roll over panel is configured to lay flat on the bottom panel and abutting the liner panel of the first side roll over panel in the set-up container, a third corner roll over panel extending outwardly from a first vertical edge of the second stacking pillar panel configured to roll over itself onto the stacking pillar panel to form a third corner stacking pillar, a third locking flap extending vertically from a second vertical edge of the second stacking pillar panel configured to lock the third corner stacking pillar together in the set-up container, and a third retaining flap extending from an inner edge of the third corner roll over panel configured to retain the third corner stacking pillar in an upright position in the set-up container, and fourth corner roll over panel extending outwardly from a third vertical edge of the second stacking pillar panel, configured to roll over itself onto the second stacking pillar panel to form a fourth corner stacking pillar. A fourth locking flap can extend laterally from a fourth vertical edge of the second stacking pillar panel configured to lock the fourth corner stacking pillar together in the set-up container, and a fourth retaining flap can extend laterally from an inner edge of the fourth corner roll over panel configured to retain the fourth corner stacking pillar in an upright position in the set-up container.

[0016] The second side roll over panel can extend laterally from a fourth edge of the bottom panel configured to roll over itself to such that the third retaining flap of the third corner roll over panel and the fourth retaining flap of the fourth corner roll over panel are sandwiched between an inner panel and an outer panel of the second side roll over panel, and liner panel of the second side roll over panel is configured to lay flat on the bottom panel abutting the liner panel of the first end roll over panel and the liner panel of the second end roll over panel in the set-up container.

[0017] In embodiments, each of the, second, third, and fourth corner roll over panels can include a respective: first inner panel, second inner panel, medial panel, first outer panel, and second outer panel. The respective first inner panel can be configured to lay flush on an inner surface of the respective first outer panel and the respective second inner panel can be configured to lay flush on an inner surface of the respective second outer panel in a respective constructed corner stacking pillar. The respective corner roll over panel can be configured to form a respective triangular stacking pillar and the respective medial panel is configured to form a hypotenuse of the respective triangular stacking pillar.

[0018] The respective medial panel of the respective corner roll over panel can include a respective locking aperture defined therein to receive a respective locking tab of a respective locking flap in the set-up container, and the respective first outer panel can include a respective stacking tab configured to be inserted into a respective aperture of a second container stacked on top of the set-up container.

[0019] The bottom panel can include a respective first and second locking aperture defined in each edge of the bottom panel configured to receive a respective first and second locking tab of each of the respective inner panels of the first end roll over panel, the first side roll over panel, the second end roll over panel, and the second ride roll over panel. The bottom panel can further include a respective first and second stacking aperture defined in the second and fourth edges of the bottom panel configured to receive a respective stacking tab of a respective corner stacking pillar of a second container when the set-up container is stacked atop the second container.

[0020] In accordance with at least one aspect of this disclosure, a display container (e.g., a container for viewing product therein without deconstructing the container), can include, a bottom wall configured to hold product within the container; a first end wall having a first end panel and a first and second stacking pillar extending upward from the bottom panel, a second end wall having a second end panel and a third and fourth stacking pillar extending upward from the bottom panel, a first sidewall having a first side panel extending between the first stacking pillar and the fourth stacking pillar, and a second sidewall having a second side panel extending between the second stacking pillar and the third stacking pillar. The first end wall, the second end wall, the first sidewall, and the second sidewall can form a product display space within the container,

[0021] In embodiments, the first, second, third, and fourth stacking pillars can be configured to provide structural stacking support to stack one or more additional containers atop the container and form viewing windows therebetween to display product within the container and without deconstructing the container.

[0022] In embodiments, the stacking pillars can be triangular stacking pillars, and a bottom edge of a hypotenuse of each of the triangular stacking pillar can be flush with the bottom panel for an entirety of a length of the bottom edge (e.g., lay flat and continuous on the bottom panel).

[0023] In embodiments, at least a portion of the triangular stacking pillars can be two-ply reinforced, and at least a portion of the bottom wall can be two-ply reinforced.

[0024] These and other features of the systems and methods of the subject disclosure will become more readily apparent to those skilled in the art from the following detailed description taken in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] So that those skilled in the art to which the subject disclosure appertains will readily understand how to make and use the devices and methods of the subject disclosure without undue experimentation, other embodiments thereof will be described in detail herein below with reference to certain figures, wherein:

[0026] FIG. **1** is a plan view of a blank in accordance with this disclosure, showing a blank for forming a container;

[0027] FIG. **2** is a perspective view of the blank of FIG. **1**;

[0028] FIGS. **3-6** show a folding sequence of forming the blank into a container, wherein:

[0029] FIG. **3** shows the formation of corner stacking pillars of the container;

[0030] FIG. **4** shows the formed corner stacking pillars;

[0031] FIG. **5** shows the erection of end walls of the container; and

[0032] FIG. **6** shows the erection of sidewalls of the container;

[0033] FIG. **7** is a perspective view of a container formed from the blank of FIG. **1** and folded using the sequence shown in FIGS. **3-6**; and

[0034] FIG. **8** is an environmental perspective view of the container of FIG. **7**, showing product held within the container, and a second container stacked atop the container formed from the blank of FIG. **1**;

[0035] FIG. **9** is a plan view of a blank in accordance with this disclosure, showing another blank for forming another container; and

[0036] FIG. **10** is a perspective view of a container formed from the blank of FIG. **9**.

DETAILED DESCRIPTION

[0037] Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, an illustrative view of an embodiment of a blank for forming a container in accordance with the disclosure is shown in FIG. **1** and is designated generally by reference character **100**. Other embodiments and/or aspects of this disclosure are shown in FIGS. **2-8**.

[0038] In accordance with at least one aspect of this disclosure, a blank **100** for forming a container **1000** can include a bottom panel **102**, and an end panel **104** extending outwardly from a first edge **106** of the bottom panel at a first bottom panel fold line **108**. The end panel **104** can include a stacking pillar panel **110** extending laterally from the first bottom panel fold line **106**, an end roll over panel **112** extending laterally from a horizontal edge **114** of the stacking pillar panel **112** at a first stacking pillar panel fold line **118**. A corner roll over panel **120** extends laterally from a first vertical edge **122** of the stacking pillar panel **112** at a first corner roll over fold line **124**, a locking flap **126** extends laterally from a second vertical edge **128** of the stacking pillar panel **112** at a locking flap fold line **130**, and retaining flap **132** extends from an inner edge **134** of the corner roll over panel **120** at a retaining flap fold line **136**.

[0039] The end roll over panel **104** includes a liner panel **138**, an inner panel **140** and an outer panel **142**, where a horizontal portion **144** of the stacking pillar panel **112** forms the outer panel **142** of the end roll over panel **104**. The end roll over panel **104** is configured to roll over the horizontal portion **144** of the stacking pillar panel **112** (e.g., the outer panel **142** of the end roll over panel **104**) at the first stacking pillar panel fold line **118** to form the inner panel **140** of the end roll over panel **104**, and the liner panel **138** is configured fold at a first end roll over panel fold line **146** such that the liner panel **138** of the end roll over panel **104** lays flat on the bottom panel **102** in a set-up container **1000**.

[0040] The corner roll over panel **120** includes, from right to left in FIG. **1**, a first inner panel **148**, a second inner panel **150**, a medial panel **152**, a first outer panel **154**, and a second outer panel **156** configured to roll over one another onto the stacking pillar panel **112** to form a corner stacking pillar **158** in the set-up container **1000**. The second outer panel **156** can be a first vertical portion **160** of the stacking pillar panel **112**. As shown, the constructed stacking pillar **158** can be triangular in shape. When the stacking pillar **158** is constructed, the first inner panel **148** lays flush on an inside surface **162** of the first outer panel **154** and the second inner panel **150** lays flush on an inner surface **164** of the second outer panel **156**. The medial panel **152** can form a hypotenuse **166** of the stacking pillar **158** when the stacking pillar **158** is constructed, e.g., as shown in FIG. **3**.

[0041] A bottom edge **168** of the medial panel **152** of the corner roll over panel **120** can lay flush on the bottom panel **102** for an entire length of the bottom edge **168** of the medial panel **152** (e.g., the bottom edge **168** is flat and continuous along the bottom panel **102** for its entire length) to provide greater structural integrity under stacking weight. For example, the continuous edge **168** can prevent the corner stacking pillars **158** from crushing and/or folding or falling inwards towards the center of the container **1000**.

[0042] The locking flap **126** includes a locking tab **170** extending laterally from the locking flap **126** at a locking flap fold line **172** configured to lock the corner stacking pillar **158** together in the set-up container **1000**, e.g., by inserting the locking flap **172** into a locking aperture **174** of the medial panel **152** of the corner roll over panel **120**. The first outer panel **154** can include a stacking tab **176** extending laterally from a top edge **178** of the first outer panel **154** configured to be inserted into a stacking aperture **180** of another container **2000** stacked on top of the set-up container **1000**. As shown in FIG. **4**, for example, the stacking tab **176** can extend upwards in the set-up container **1000**. In embodiments, the stacking tab **176** can be closer to an inner side corner **182** of the stacking pillar **158** than an outer corner **184** of the stacking pillar **158**.

[0043] As can be seen in FIGS. **5**, the first inner panel **148** and the first outer panel **154** of the corner roll over panel **120** can form a first two ply side **186** of the constructed stacking pillar **158** and the second inner panel **150** and the second outer panel **156** of the corner roll over panel **120** can form a second two ply side **188** of the constructed stacking pillar **158** to reinforce a stacking strength of the constructed stacking pillar **158** relative to a direction of gravity. Further, as shown, the direction of the corrugation flutes can be orientated parallel to the direction of the stacking forces to provide stacking strength and reduce the risk of collapse of the stacking pillars **158** under the weight of additional containers.

[0044] The retaining flap **132** extends from the inner edge **134** of the corner roll over panel **120** at the retaining flap fold line **136** and is separated from the first outer panel **154** by a cut line **190**. The retaining flap **132** is configured to retain the corner stacking pillar **158** in an upright position in the set-up container **1000** by sandwiching the retaining panel **132** within a side roll over panel **201**, e.g., as described further below. The retaining flap can also include finger cut outs **192** to assist a user in moving the retaining flaps to the extended position, e.g., as shown in FIGS. **1-3**.

[0045] The side roll over panel **201** extends outwardly from a second edge **203** of the bottom panel **102** at a second bottom panel fold line **205**. The side roll **201** over panel includes a liner panel **207**, an inner panel **209** and an outer panel **211**, where the inner panel **209** extends between a first **213** and second **215** side roll over panel fold lines. The side roll over panel **201** is configured to roll over itself, were the outer panel **211** rolls up at the second bottom panel fold line **205** to an upright position, the inner panel **209** rolls over the outer panel at the first side roll over panel fold line **213**, and the liner panel **207** rolls over the inner panel **209**, folding at the second side roll over panel fold line **205**, and over the retaining flap **132** of the corner roll over panel **120**. The retaining panel **132** is thus sandwiched between the inner panel **209** and the outer panel **211** of the side roll over panel **201** in the set-up container **1000**, and the liner panel **207** of the side roll over panel **201** lays flat on the bottom panel **102** in the set-up container **1000**. A corner **217** of the liner panel **207** of the side roll over panel **201** abuts the liner panel **138** of the end roll over panel **112** in the set-up container **1000**, for example, as shown in FIG. **7**. In embodiments, the corner **217** of the liner panel **207** can be cut out to receive the corner **139** of the liner panel of the end roll over panel **112** in the set-up container so that the liner panels **138** and **207** can "lock" together.

[0046] In embodiments, the bottom panel **102** can include a first locking aperture **219** and a second locking aperture **221** defined in the first edge **106** of the bottom panel **102**, and the inner panel **140** of the end roll over panel **112** can include a first locking tab **223** and a second locking tab **225** configured to be inserted into the first locking aperture **219** and the second locking aperture **221**, respectively, of the bottom panel **102** in the set-up container **1000**. The bottom panel **102** can also include a third locking aperture **227** and a fourth locking aperture **229** defined in the second edge **203** of the bottom panel **102** and the inner panel **209** of the side roll over panel **201** can include a first locking tab **231** and a second locking tab **233** configured to be inserted into the third locking aperture **227** and the fourth locking aperture **229**, respectively, of the bottom panel **102** in the set-up container **1000**.

[0047] In embodiments, the bottom panel **102** can further include a fifth locking aperture **235** and a sixth locking aperture **237** defined in the second edge **203** of the bottom panel configured to receive a respective stacking tab **276** of a second container **2000** when the set-up container **1000** is stacked atop the second container **2000**.

[0048] The corner roll over panel as described herein above can be a first roll over corner panel **120**, and the end panel **104** can further include a second corner roll over panel extending laterally from a third vertical edge of the stacking pillar panel **110**, a second locking flap **226** extending laterally from a fourth vertical edge **228** of the stacking pillar panel **110** configured to lock the second corner stacking pillar **258** together in the set-up container **1000**, and a second retaining flap **232** extending laterally from an inner edge **234** of the second corner roll over panel **220** configured to retain the second corner stacking pillar **258** in an upright position in the set up container **1000**.

[0049] In embodiments, a second end panel **204** extending from a third edge **206** of the bottom panel **102** and can include a second stacking pillar panel **210**, a second end roll over panel **212**, and a third and **320** a fourth corner roll over panel **420**. The respective portions and/or panels of the second end panel **204** can be the same or similar to that described with respect to the first end panel **104**, and as such common elements already described with respect to the first end panel **104** are not repeated herein for the second end panel **204**. Similarly, the blank **100** can include a second side roll over panel **501** extending from a fourth edge **503** of the bottom panel, and the second side roll over panel **501** can be similar to the first side roll over panel **201**, thus common elements between the first and second side roll over panels **201**, **501** already described for the first side roll over panel **201** are not repeated with respect to the second side roll over panel **501**.

[0050] In embodiments, each of the first **120**, second **220**, third **320**, and fourth **420** corner roll over panels can include a respective: first inner panel **148**, **248**, **348**, **448**, second inner panel **150**, **250**, **350**, **450**, medial panel **152**, **252**, **352**, **452**, first outer panel, **154**, **254**, **354**, **454**, and second outer panel **156**, **256**, **356**, **456**. Each stacking pillar **158**, **258**, **358**, **458** is held together by a respective locking flap **126**, **226**, **326**, **426**, and the end panels **104**, **204** are held upright with respective retaining flaps **132**, **232**, **332**, **432**.

[0051] In accordance with at least one aspect of this disclosure, and as shown in FIGS. **7-8**, a display container **1000** (e.g., a container for viewing product therein without deconstructing the container), can include a bottom wall **1002** configured to hold product **1051** within the container **1000**. From the bottom wall **1102** extending upward, the container includes a first end wall **1104** having a first end panel **104** and a first **1158** and second stacking pillar **1058** and a second end wall **1204** having a second end panel and a third **1358** and fourth **1458** stacking pillar extending upward from the bottom wall **1102**. Also extending upward from the bottom wall **1102**, the container includes a first sidewall **1201** having a first side panel **1209** extending between the first stacking pillar **1158** and the fourth stacking pillar **1458**, and a second sidewall **1501** having a second side panel **1509** extending between the second stacking pillar **1258** and the third stacking pillar **1358**. The first end wall **1104**, the second end wall **1204**, the first sidewall **1201**, and the second sidewall **1501** can form a product display space **1053** within the container **1000** and within the stacking pillars **1158**, **1258**, **1358**, **1458**. One or more portions of the container **1000**, e.g., the stacking pillars **1158**, **1258**, **1358**, **1458**, the side walls **1201**, **1508**, the end walls **1104**, **1204**, and the bottom wall **1102** can be 2-ply reinforced to provide structural stacking support to be able to stack one or more additional containers **2000** atop the container **1000**, e.g., as shown in FIG. **8**.

[0052] When stacked, the stacking pillars **1158**, **1258**, **1358**, **1458** can form viewing windows therebetween to display product **1051** within the container **1000**. This way, the product can be displayed on the shelves within the original containers **1000** and viewed from all sides, while the containers **1000**, **2000** remain stacked, and without the need to deconstruct the container **1000**.

[0053] As shown in FIGS. **2-7**, a folding sequence to erect the container **1000** from the blank **100** can be as follows. From the flat blank **100**, e.g., as shown in FIG. **2**, the stacking pillars **1158**, **1258**, **1358**, **1458** are formed by rolling the corner roll over panels **120**, **220**, **320**, **420**, starting at the respective first inner panel **148**, **248**, **348**, **448** and rolling inwards towards the respective stacking pillar panel **110**, **210** so that the respective second inner panel **150**, **250**, **350**, **450** lays flat on the inner surface of the stacking pillar panel **110**, **210**. As can be seen in FIG. **7**, the first inner panel **148**, **248**, **348**, **448** and the first outer panel **154**, **254**, **354**, **454** can form a respective first two ply section **186**, **286**, **386**, **486** of the respective stacking pillars **1158**, **1258**, **1358**, **1458**. The respective second inner panel **150**, **250**, **350**, **450** and the respective second outer panel **156**, **256**, **356**, **456** form a respective second two ply section of the respective stacking pillar **1158**, **1258**, **1358**, **1458**. The respective locking flaps **126**, **226**, **326**, **426** are folded over the respective locking flap fold lines **130**, **230**, **330**, **430** and the respective locking tab **170**, **270**, **370**, **470** is inserted into the respective locking aperture **174**, **274**, **374**, **474** of the medial panel of the stacking pillar. This forms an overlapping two-ply portion on the respective medial panels.

[0054] As is seen in FIG. **3**, once the corner roll over panels **120**, **220**, **320**, **420** are "rolled" and the stacking pillars **1158**, **1258**, **1358**, **1458** are formed, the retaining flaps **132**, **232**, **332**, **432** extend upright, and are not rolled with the corner roll over panels **120**, **220**, **320**, **420**. The end panels **104**, **204** are then folded upwards at the first bottom panel fold lines to erect the end walls **1104**, **1204**, into an upright position. Next, the end roll over panels **112**, **212** are folded over the first end roll over panel fold line, and the respective locking tabs (e.g., **223**, **225**) of the inner panel of the end roll over panels are inserted into the respective first and second locking apertures (e.g., **219**, **221**) of the bottom panel **102**. Then, the respective liner panel **138**, **238** of the end roll over panels is placed onto the bottom panel **102**. Locking the end roll over panels **112**, **212** in place will retain the end panels in their upright position as the side walls **1201**, **1501** are formed.

[0055] After the first and second end walls **1104**, **1204** are locked in place, the sidewalls **1201**, **1501** can be formed from the side roll over panels **201**, **501**. First, the outer panel **211**, **511** is folded upward at the second bottom panel fold lines **205**, **505**, and the inner panels **209**, **509** are rolled over the first side roll over panel fold lines **213**, **513** to lay against the inner surface of the respective outer panels **211**, **511**, and the respective locking tabs (e.g., **231**, **233**) are inserted into the respective third and fourth locking apertures (e.g., **227**, **229**) in the second edge **203** and fourth edge **503** of the bottom panel **102**. While the side roll over panels **201**, **501** are being rolled over, the retaining flaps **132**, **232**, **332**, **432** of the now upright respective stacking pillars **1158**, **1258**, **1358**, **1458** extend horizontally along the second and fourth bottom edges **203**, **503** of the bottom panel **102**. Thus, when the respective inner panel **209**, **509** of the side roll over panel is rolled and locked into the bottom panel **102**, the respective retaining flap is caught between the inner **209**, **509** and outer panels **211**, **511** of the side roll over panels **201**, **501** and sandwiched therebetween, to add additional support for holding the stacking pillars upright **1158**, **1258**, **1358**, **1458**, beyond the locking tabs of the end roll over panel. This configuration and folding sequence allow for a single user to manually fold the blank into the container, and allows for doing so without any adhesive.

[0056] As shown in FIGS. **9** and **10**, a blank **500** for constructing a container **5000**, can be similar to that of blank **100** for constructing container **1000**, for example blank **500** can have similar components and features, and may be erected in a similar manner to that described with respect to blank **100** and container **1000**. For brevity, the description of common elements that have been described above are not repeated with respect to FIGS. **9** and **10**. In blank **500**, the stacking tabs can be define along the corners and can be two plies, effectively forming two stacking tabs **576*a*** and **576*b***. For example, as shown with respect to corner roll over panel **520** in FIG. **9**, a first portion of a first ply of the stacking tab **576*a*** can be defined on the first inner panel **548** and a second portion of stacking tab **576*a*** of the first ply can be defined on the second inner panel **550**. A first portion of a second ply of the stacking tab **576*b*** can be defined on the first outer **554** and a second portion of stacking tab **576*b*** of the second ply can be defined on the second inner panel **556**. As shown in FIG. **10**, the stacking tab **576*a*** forms the inner (first) ply while stacking tab **576*b*** forms the outer (second) ply. While the stacking tabs **576*a***, **576*b*** are described with respect to the corner roll over panel **520**, it will be appreciated that each of the corner roll over panels and stacking pillars shown in FIGS. **9** and **10** can be the same as shown for corner roll over panel **520**.

[0057] Further, as seen in FIGS. **9** and **10**, the outer panel **542** of the end roll over panel **504**, and the outer panel **611** of the side roll over panel **601** each have a cut out portion forming a corner cut out **569**. In the set up container **5000**, the as shown in the corner roll over panel **520**, the corner cut out **569** forms a single ply portion in the corner roll over panel **520** to accept a stacking tab **576*a***, **576*b*** of a second set up container stacked below the container **5000**. While the corner cut out **569** is described with respect only to one corner, e.g., as formed on the corner roll over panel **520** on the set up container **5000**, it will be appreciated that each corner can include a corner cut out **569** as shown for corner roll over panel **520**.

[0058] Those having ordinary skill in the art understand that any numerical values disclosed herein can be exact values or can be values within a range. Further, any terms of approximation (e.g., "about", "approximately", "around") used in this disclosure can mean the stated value within a range. For example, in certain embodiments, the range can be within (plus or minus) 20%, or within 10%, or within 5%, or within 2%, or within any other suitable percentage or number as appreciated by those having ordinary skill in the art (e.g., for known tolerance limits or error ranges).

[0059] The articles "a", "an", and "the" as used herein and in the appended claims are used herein to refer to one or to more than one (i.e., to at least one) of the grammatical object of the article unless the context clearly indicates otherwise. By way of example, "an element" means one element or more than one element.

[0060] The phrase "and/or," as used herein in the specification and in the claims, should be understood to mean "either or both" of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with "and/or" should be construed in the same fashion, i.e., "one or more" of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the "and/or" clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to "A and/or B", when used in conjunction with open-ended language such as "comprising" can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0061] As used herein in the specification and in the claims, "or" should be understood to have the same meaning as "and/or" as defined above. For example, when separating items in a list, "or" or "and/or" shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as "only one of" or "exactly one of," or, when used in the claims, "consisting of," will refer to the inclusion of exactly one element of a number or list of elements. In general, the term "or" as used herein shall only be interpreted as indicating exclusive alternatives (i.e., "one or the other but not both") when preceded by terms of exclusivity, such as "either," "one of," "only one of," or "exactly one of."

[0062] Any suitable combination(s) of any disclosed embodiments and/or any suitable portion(s) thereof are contemplated herein as appreciated by those having ordinary skill in the art in view of this disclosure.

[0063] The embodiments of the present disclosure, as described above and shown in the drawings, provide for improvement in the art to which they pertain. While the apparatus and methods of the subject disclosure have been shown and described, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the scope of the subject disclosure.

What is claimed is:

1. A blank for forming a container, comprising:

a bottom panel;

an end panel extending outwardly from a first edge of the bottom panel, wherein the end panel includes:

a stacking pillar panel extending laterally from a first edge of the bottom panel;

an end roll over panel extending laterally from a horizontal edge of the stacking pillar panel configured to roll over the stacking pillar panel such that a liner panel of the end roll over panel is configured to lay flat on the bottom panel in a set-up container;

a corner roll over panel extending laterally from a first vertical edge of the stacking pillar panel configured to roll over itself onto the stacking pillar panel to form a corner stacking pillar;

a locking flap extending a laterally from a second vertical edge of the stacking pillar panel configured to lock the corner stacking pillar together in the set-up container; and

a retaining flap extending from an inner edge of the corner roll over panel configured to retain the corner stacking pillar in an upright position in the set-up container; and

a side roll over panel extending outwardly from a second edge of the bottom panel configured to roll over itself to such that the retaining flap of the corner roll over panel is sandwiched between an inner panel and an outer panel of the side roll over panel, and wherein a liner panel of the side roll over panel is configured to lay flat on the bottom panel abutting the liner panel of the end roll over panel in the set-up container.

2. The blank of claim **1**, wherein the stacking pillar is a triangular stacking pillar.

3. The blank of claim **2**, wherein the corner roll over panel includes, a first inner panel, a second inner panel, a medial panel, a first outer panel, and a second outer panel, wherein the first inner panel is configured to lay flush on an inside surface of the first outer panel, wherein the second inner panel is configured to lay flush on an inner surface of the second outer panel with the corner stacking pillar constructed, wherein the medial panel is configured to form a hypotenuse of the stacking pillar in a constructed stacking pillar.

4. The blank of claim **3**, wherein the first inner panel and the first outer panel are configured to form a first two ply side of the constructed stacking pillar and wherein the second inner panel and the second outer panel are configured to form a second two ply side of the constructed stacking pillar to reinforce a stacking strength of the constructed stacking pillar relative to a direction of gravity.

5. The blank of claim **4**, wherein a bottom edge of the medial panel of the corner roll over panel is configured to lay flush on the bottom panel for an entire length of the bottom edge of the medial panel.

6. The blank of claim **5**, wherein the medial panel of the corner roll over panel includes a locking aperture defined therein to received a locking tab of the locking flap in a constructed triangular stacking pillar.

7. The blank of claim **6**, wherein the first outer panel includes a stacking tab extending laterally from a top edge of the first outer panel configured to be inserted into a stacking aperture of another container stacked on top of the set-up container.

8. The blank of claim **7**, wherein the bottom panel includes a first locking aperture and a second locking aperture defined in the first edge of the bottom panel, wherein the inner panel of the end roll over panel includes a first locking tab and a second locking tab configured to be inserted into the first locking aperture and the second locking aperture, respectively, of the bottom panel in the set-up container.

9. The blank of claim **8**, wherein the bottom panel includes a third locking aperture and a fourth locking aperture defined in the second edge of the bottom panel, wherein the inner panel of the side roll over panel includes a first locking tab and a second locking tab configured to be inserted into the third locking aperture and the fourth locking aperture, respectively, of the bottom panel in the set-up container.

10. The blank of claim **9**, wherein the bottom panel further includes a fifth locking aperture and a sixth locking aperture defined in the second edge of the bottom panel configured to receive a respective stacking tab of a second container when the set-up container is stacked atop the second container.

11. The blank of claim **10**, wherein the corner roll over panel is a first roll over corner panel, and wherein the end panel further includes:

a second corner roll over panel extending laterally from a third vertical edge of the stacking pillar panel, configured to roll over itself onto the stacking pillar panel to form a second corner stacking pillar;

a second locking flap extending laterally from a fourth vertical edge of the stacking pillar panel configured to lock the second corner stacking pillar together in the set-up container; and

a second retaining flap extending laterally from an inner edge of the second corner roll over panel configured to retain the second corner stacking pillar in an upright position in the set-up container, wherein the side roll over panel is configured to sandwich the second retaining panel between the inner panel and the outer panel of the side roll over panel in the set-up container.

12. The blank of claim **11**, wherein the side roll over panel is a first side roll over panel and, wherein the end panel is a first end panel, and further comprising a second end panel extending outwardly from a third edge of the bottom panel, wherein the second end panel includes:

a second stacking pillar panel extending outwardly from the third edge of the bottom panel;

a second end roll over panel extending laterally from a horizontal edge of the second stacking pillar panel configured to roll over the stacking pillar panel such that a liner panel of the second end roll over panel is configured to lay flat on the bottom panel and abutting the liner panel of the first side roll over panel in the set-up container;

a third corner roll over panel extending outwardly from a first vertical edge of the second stacking pillar panel configured to roll over itself onto the stacking pillar panel to form a third corner stacking pillar;

a third locking flap extending vertically from a second vertical edge of the second stacking pillar panel configured to lock the third corner stacking pillar together in the set-up container; and

a third retaining flap extending from an inner edge of the third corner roll over panel configured to retain the third corner stacking pillar in an upright position in the set-up container; and

a fourth corner roll over panel extending outwardly from a third vertical edge of the second stacking pillar panel, configured to roll over itself onto the second stacking pillar panel to form a fourth corner stacking pillar,

a fourth locking flap extending laterally from a fourth vertical edge of the second stacking pillar panel configured to lock the fourth corner stacking pillar together in the set-up container; and

a fourth retaining flap extending laterally from an inner edge of the fourth corner roll over panel configured to retain the fourth corner stacking pillar in an upright position in the set-up container.

13. The blank of claim **12**, further comprising the second side roll over panel extending laterally from a fourth edge of the bottom panel configured to roll over itself to such that the third retaining flap of the third corner roll over panel and the fourth retaining flap of the fourth corner roll over panel are sandwiched between an inner panel and an outer panel of the second side roll over panel, and liner panel of the second side roll over panel is configured to lay flat on the bottom panel abutting the liner panel of the first end roll over panel and the liner panel of the second end roll over panel in the set-up container.

14. The blank of claim **13**, wherein each of the, second, third, and fourth corner roll over panels include a respective:

first inner panel, second inner panel, medial panel, first outer panel, and second outer panel, wherein the respective first inner panel is configured to lay flush on an inner surface of the respective first outer panel, wherein the respective second inner panel is configured to lay flush on an inner surface of the respective second outer panel in a respective constructed corner stacking pillar, wherein the respective corner roll over panel is configured to form a respective triangular stacking pillar, wherein the respective medial panel is configured to form a hypotenuse of the respective triangular stacking pillar.

15. The blank of claim **14**, wherein the respective medial panel of the respective corner roll over panel includes a respective locking aperture defined therein to receive a respective locking tab of a respective locking flap in the set-up container, and wherein the respective first outer panel includes a respective stacking tab configured to be inserted into a respective aperture of a second container stacked on top of the set-up container.

16. The blank of claim **15**, wherein the bottom panel includes a respective first and second locking aperture defined in each edge of the bottom panel configured to receive a respective first and second locking tab of each of the respective inner panels of the first end roll over panel, the first side roll over panel, the second end roll over panel, and the second ride roll over panel.

17. The blank of claim **16**, wherein the bottom panel further includes a respective first and second stacking aperture defined in the second and fourth edges of the bottom panel configured to receive a respective stacking tab of a respective corner stacking pillar of a second container when the set-up container is stacked atop the second container.

18. A display container, comprising:

a bottom wall configured to hold product within the container;

a first end wall having a first end panel and a first and second stacking pillar extending upward from the bottom panel;

a second end wall having a second end panel and a third and fourth stacking pillar extending upward from the bottom panel;

a first sidewall having a first side panel extending between the first stacking pillar and the fourth stacking pillar; and

a second sidewall having a second side panel extending between the second stacking pillar and the third stacking pillar, wherein the first end wall, the second end wall, the first sidewall, and the second sidewall form a product display space within the container,

wherein the first, second, third, and fourth stacking pillars are configured to provide structural stacking support to stack one or more additional containers atop the container and form viewing windows therebetween to display product within the container and without deconstructing the container.

19. The display container of claim **18**, wherein the stacking pillars are triangular stacking pillars, and wherein a bottom edge of a hypotenuse of each of the triangular stacking pillar is flush with the bottom panel for an entirety of a length of the bottom edge.

20. The display container of claim **19**, wherein at least a portion of the triangular stacking pillars are two-ply reinforced, and wherein at least a portion of the bottom wall is two-ply reinforced.

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